Enhancing polyp detection in endoscopy images

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***Abstract*—** Digital image processing plays a critical role in advancing polyp detection in gastrointestinal endoscopy, supporting clinicians in accurate diagnosis and timely intervention. While deep learning-based segmentation methods like U-Net have shown promise, challenges remain due to variability in polyp appearance, small sizes, and low contrast with surrounding tissues. In this work, we enhance the U-Net framework through multiple architectural improvements and training strategies. We evaluate Deep U-Net, Attention U-Net, and pretrained encoder-based models (MobileNetV2 and ResNet50), and integrating data augmentation. Attention mechanisms are employed to selectively enhance relevant features and suppress background noise. To better handle low-confidence cases, we optimize training using a weighted combination of Dice and IoU losses. Our experiments show that the Attention U-Net significantly outperforms other models, achieving up to 22% improvement in mDice and 19% in mIoU, with strong gains on the most challenging test samples. These results highlight the effectiveness of attention-enhanced architectures and robust training strategies in building more reliable polyp segmentation systems.

Keywords—polyp segmentation, U-Net, Digital image processing,

I. Introduction

Colorectal cancer ranks among the most prevalent cancers worldwide, with polyps—abnormal growths on the inner lining of the colon or rectum—often serving as precursors to this disease. Early detection and removal of these polyps are crucial in preventing their progression to malignancy.[1] However, during colonoscopy examinations, polyps can be missed due to their subtle appearance or challenging locations, leading to miss rates between 14% and 30% depending on their type and size. To enhance detection rates and reduce the incidence of colorectal cancer, the development of automatic, computer-aided detection systems has become imperative.[2]

The Kvasir-SEG dataset addresses this need by providing an open-access collection of 1,000 gastrointestinal polyp images, each accompanied by a manually annotated segmentation mask verified by experienced gastroenterologists. The images are RGB and vary in resolution from 332×487 to 1920×1072 pixels and are stored alongside their corresponding masks, facilitating research in polyp segmentation and automatic analysis of colonoscopy images. This dataset serves as a valuable resource for developing and benchmarking algorithms aimed at improving polyp detection and segmentation. [3]

In recent years, several studies have proposed methods to enhance polyp detection and segmentation. These methods mostly use cutting-edge deep learning approaches, such as a variety of pre-trained encoders and custom decoder architectures [4,5], and have introduced lightweight networks, such as NanoNet, for real-time detection [6]. However, many of these methods are computationally expensive.

In this project, we implemented lightweight U-Net-based segmentation models and trained them on 70% of the Kvasir-SEG dataset, using 15% for evaluation and 15% for the test. We experimented with various input configurations, including original-resolution images, downsampled inputs, and images enhanced through Gaussian filtering, histogram equalization, and morphological operations. Additionally, we explored data augmentation techniques and investigated more advanced architectures such as Deep U-Net and Attention U-Net. We also integrated pretrained encoder backbones like MobileNetV2 and ResNet50 to evaluate the impact of transfer learning on segmentation performance.

II. Method

we use the U-Net architecture for polyp detection due to its ability to capture fine-grained details and provide accurate segmentations. The model consists of three layers: a contracting path (encoder), a bottleneck, and an expanding path (decoder). The encoder progressively reduces spatial dimensions through convolutional layers and max pooling, capturing context from the input image. The bottleneck is the point of lowest resolution, where the feature maps are most compact, allowing the network to capture the most important features. The decoder then upscales the feature maps using transposed convolutions and skip connections from the encoder, restoring spatial resolution while maintaining crucial details for accurate segmentation.

The network uses 3x3 convolutional kernels to extract features and applies ReLU activation for non-linearity, with dropout regularization to prevent overfitting. The final layer is a softmax activation for multi-class classification, followed by a pixel classification layer to assign a label to each pixel. The model has 7.6 million learnable parameters, a learning rate of 1e-4, a maximum of 10 epochs for training, and a mini-batch size of 4. These configurations ensure that the network learns effectively while managing computational complexity.

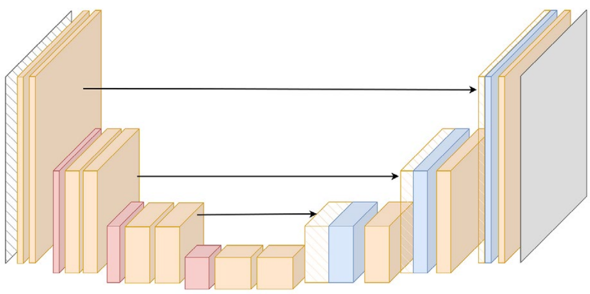

Fig. 1. an example of U-Net Architecture[7]

First, we trained the models with different inputs resulting in different models called: A. U-Net_origin, B. U_Net_Resized C. U_Net_Enhanced D. U_Net_MO_processed. In the following sections we briefly describe of each.

A. U-Net_origin,

For this model, the input size was resized to 1072×1344 resolution, So that it becomes dividable by 2^3 (3 number of U_Net layers). The training process took 876 minutes for 9 epochs out of 20. Expected to preserve details.

B. U-Net_Resized

The input of this model was resized and downsampled to 64×64 resolution, this one is also dividable by 2^3 . But expected to have lower runtime with comparable results.

C. U_Net_Enhanced

Here, we first apply Gaussian filter with standard deviation of 4 to smooth it and then equalize the histogram for each channel. A preprocessed image is shown in figure2.

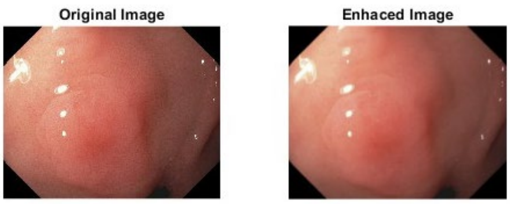

Fig. 2. Preprocessing for U-Net Enhanced model, Left: Original image Right: Enhanced image using gaussian filter and equalizing RGB histograms.

D. U-Net_MO_processed

For these models we applied four different morphological operations on the images and for each trained the U-Net model, below we explain each operations, Here we have RGB images, so morphological operations do not apply directly as in grayscale or binary images. Instead, these operations are performed independently on each color channel (Red, Green, and Blue).

Dilation: Expands bright regions in an image. For RGB images, dilation is applied separately to each channel. For each channel A and structuring element B(here is a disk with radus equals to 5), is given by

$$A \oplus B = \{z \mid (B)_z \cap A \neq \emptyset\} \quad (1)$$

The pixel value at z is the maximum intensity in the neighborhood defined by B. it Enhances object boundaries by increasing the intensity of pixels near edges.

Erosion: Shrinks bright regions by removing pixels from object boundaries.

$$A \ominus B = \{z \mid (B)_z \subseteq A\} \quad (2)$$

the pixel value at z is the minimum intensity in the neighborhood defined by B. so,it reduces the size of bright regions in each channel and can remove small noise but may also shrink important image details.

Opening: Removes small bright noise while preserving larger objects.

Defined as erosion followed by dilation.

Closing: Fills small dark holes in bright objects. Defined as dilation followed by erosion, Closes small gaps in objects.

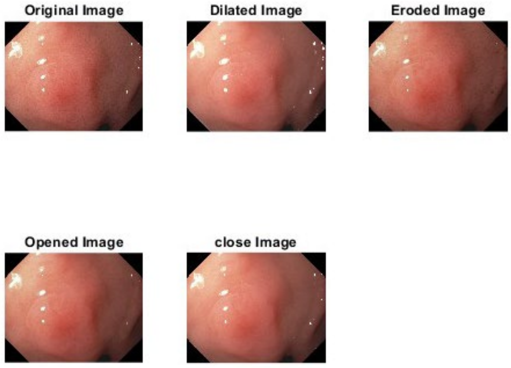

Fig. 3. Illustratres Morphological Operations on a sample image.

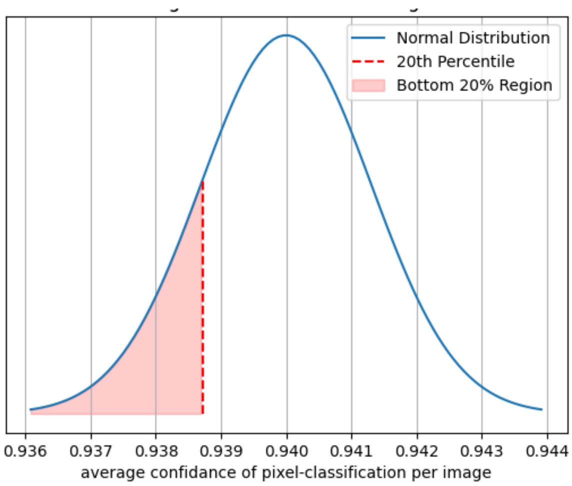

Fig. 4. Distribution of average confidance of test images in resized model

To address the missing polyps issue, we found the first 20% of least confident(figure 4. Illustrate this) to detect polyps in our test dataset based on the resized model, and then try to keep track of the performance metric of these ones specifically as well as all of the test set and see how our proposed model will improve the segmentation of these polyps.

Since the region of interest is the polyp (foreground) and not the background, it is not a good idea to use a loss function during training that weights each pixel equally. Therefore, we tried the original model with a loss function based on Intersection over Union (IOU) metric, which measures the area of overlap of the mask and predicted mask over the area of union of both masks, and also the Dice metric, which measures twice the area of overlap over the

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total area. Thus, we trained the resized model using different loss functions: 1. Binary Cross Entropy (BCE) loss, 2. Dice loss (1 - Dice coefficient), 3. BCE and Dice weighted loss, with weights of 0.3 and 0.7 respectively, 4. IOU loss (1 - IOU coefficient), and 5. IOU and Dice weighted loss, with weights of 0.75 and 0.25 respectively.

We have only 1000 images, using 700 for training, 150 for validation, and 150 for the test set. To address the low training set size issue and improve the model's performance on the bottom 20% of polyps, we augmented the data randomly using 4 different transformations: 1. Horizontal flip, 2. Vertical flip, 3. Random rotation, and 4. Adding random brightness and contrast. Figure 5 shows these transformations for one random image and its corresponding mask. Finally, we combined the augmented training data with the original training set, totaling 1400 images, to train our model with the best-performing loss function.

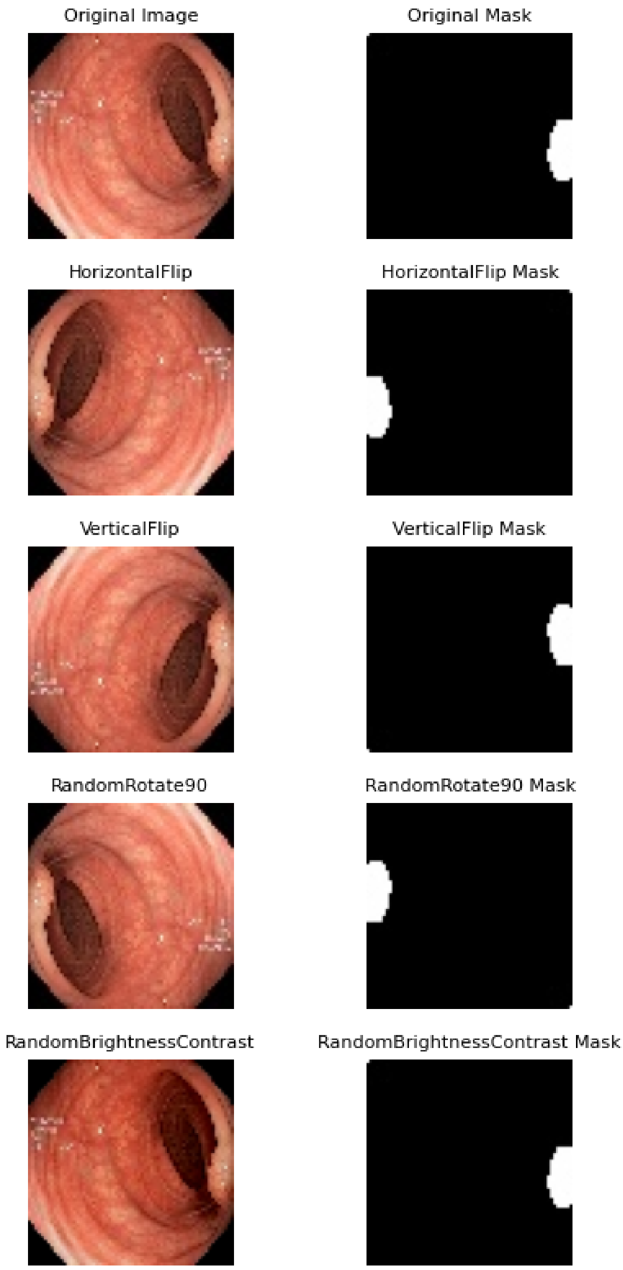

Fig. 5. Different transformation that been applied on the trainig set

We also tried different architectures other than the simple UNET, 1. Deep UNET 3. Attention U-Net 4. Fine-tuned pre-trained models.

The deep U_net has architecture shown in figure 6. Including encoder layers with double conv2D with 64, 128, 256, and 512 filters respectively and a conv2D with 1024 filters as a bottleneck, the decoder includes the skip connections and up sampling, as normal UNet would have.

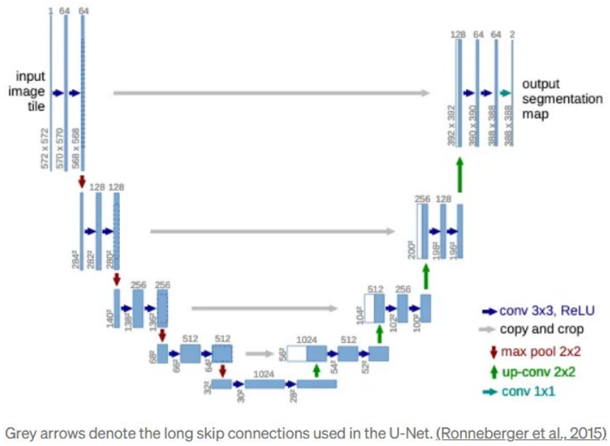

Fig. 6. The used deep Unet architecture.

In attention UNet model(shown in figure 7), **Attention Gates (AGs)** are introduced to refine the skip connections between the encoder and decoder pathways. These gates help the network focus on the most relevant features while suppressing irrelevant or noisy regions, which is particularly beneficial for tasks like medical image segmentation. Each AG receives two inputs: a gating signal from the decoder and feature maps from the encoder. These inputs are processed through a series of 1x1x1 convolutions, followed by element-wise addition, ReLU activation, another convolution, and a sigmoid function to produce attention coefficients (shown in figure 8).

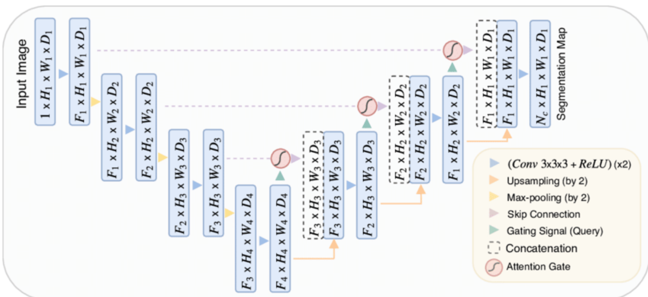

Fig. 7. Attention Unet Architecture[8]

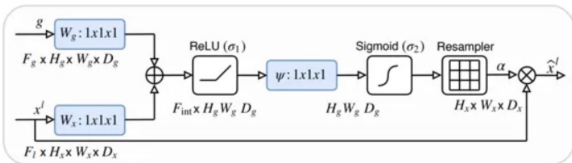

Fig. 8. Attention Gate[8]

For the fine-tuned pre-trained models, I used 2 different back bone models first MobilNet , and second ResNet. These models built upon the pretrained MobileNetV2 and ResNet50 encoder respectively. MobileNetV2, is a lightweight convolutional neural network trained on ImageNet, serves as the backbone for feature extraction due to its strong representational power and computational efficiency. Feature maps are extracted from multiple depth levels of the encoder and passed into a decoder composed of transposed convolution layers for upsampling. To enhance the model's focus on important regions, Attention Gates are inserted between corresponding encoder and decoder levels. These gates selectively highlight relevant features while suppressing less informative ones, enabling more accurate spatial localization. The combination of pretrained MobileNetV2 and attention-based skip connections allows the model to achieve high segmentation performance even with limited training data and computational resources.

In addition to the MobileNetV2-based model, I implemented a similar Attention U-Net architecture using ResNet50 as the encoder backbone. ResNet50 is a 50-layer deep convolutional neural network pretrained on ImageNet, known for its use of residual connections that allow deeper networks to train effectively by mitigating vanishing gradient issues. In this architecture, feature maps are extracted from intermediate layers of the ResNet50 encoder to preserve spatial resolution and semantic richness. These features are then passed through a decoder that includes transposed convolution layers for upsampling and attention gates that selectively highlight informative regions before concatenation. This combination leverages the strong feature extraction of ResNet50 with the spatial focus of attention mechanisms, resulting in a lightweight yet powerful segmentation model tailored for small input sizes like ours 64×64 pixels.

III. RESULTS

To train our models, we shuffled and split the dataset to 70% training and 15% validation and 15% test set.

We trained each model on 10 epochs and stopped the learning process when the training and validation loss and accuracy converged to a number. Figure 6 shows how the learning curves look after training, this figure is for U-Net-resized but for others the trend the curve is following is pretty much the same.

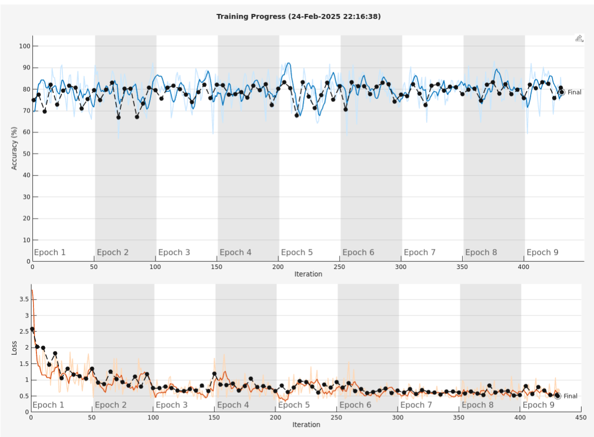

Fig. 9. Accuracy during training process on top, black dots are valiations and blue are trainset results. Loss values on bottom, orange line training and black dots validation values.

Performance of each model is mentioned in Table I, as well as the mean and standard deviation of the average confidence of prediction. Also one of the test figures have been shown, with the ground truth mask and other models prediction in figure 7. Also, the performance of the 20% least confident test set. Table II

TABLE I. MODELS PERFORMANCE FOR PROCESSED IMAGES ON ALL TEST SET

Models	**Metrics**				
	Acc	***mIOU***	***mDice***	***Conf mean***	***Conf std***
U-Net_resized	78.61 %	14.9 %	25.0 %	0.94	0.016
U-Net_enhanced	70.70%	17 %	28 %	-	-
U-Net_MO_Dilation	76.62%	18.1%	28.7 %	0.89	0.025
U-Net_MO_Erosion	76.59%	9.93%	17.3 %	0.93	0.016
U-Net_MO_Opening	77.02%	21.9 %	33.18 %	0.89	0.032
U-Net_MO_closing	75.50%	20.87%	32.12 %	0.90	0.026

TABLE II. MODELS PERFORMANCE FOR PROCESSED IMAGES ON 20% LEAST CONFIDENT TEST SET

Models	**Metrics**				
	Acc	***mIOU***	***mDice***	***Conf mean***	***Conf std***
U-Net_resized	13.30 %	3.4 %	5.4 %	0.88	0.022
U-Net_enhanced	-	-	-	-	-
U-Net_MO_Dilation	13.34 %	3.5%	5.7%	0.88	0.018
U-Net_MO_Erosion	13.86%	3.7%	5.8%	0.89	0.020
U-Net_MO_Opening	14.25%	2.3 %	3.9%	0.87	0.015
U-Net_MO_closing	14.-3%	2.3 %	3.9 %	0.87	0.018

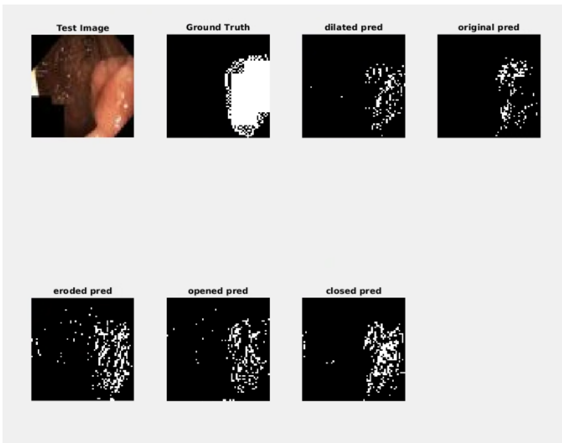

Fig. 10. One sample of the test image along side the ground truth mask and followed by models prediction

The Performance of the resized model varies using different loss function and it seems that using combination of losses with proper weights makes the performance of the model better for segmentation task, these performances has been shown in Table III. . Figure 8 shows several random samples of test images with their masks along side the predicted mask by the model with the weighted DICE and IOU losses.

TABLE III. RESIZED MODEL PERFORMANCE USING DIFFERENT LOSS FUNCTIONS FOR ALL TEST SET

Loss functions	**Metrics**				
	Acct	***mIOU***	***mDice***	***Conf mean***	***Conf std***
BCE	81.6 %	24%	33 %		
DICE	81.9%	40.4%	53.6 %		
BCE +DICE	83.5 %	44.4%	57.8 %		
IOU	84.30%	44.64%	57.59 %		
IOU (0.75)+DICE(0.25)	84.2%	48.1%	61.04 %	0.94	0.039
IOU(0.5)+DICE(0 .25) + BCE (0.25)	81%	44.79%	58.61 %		

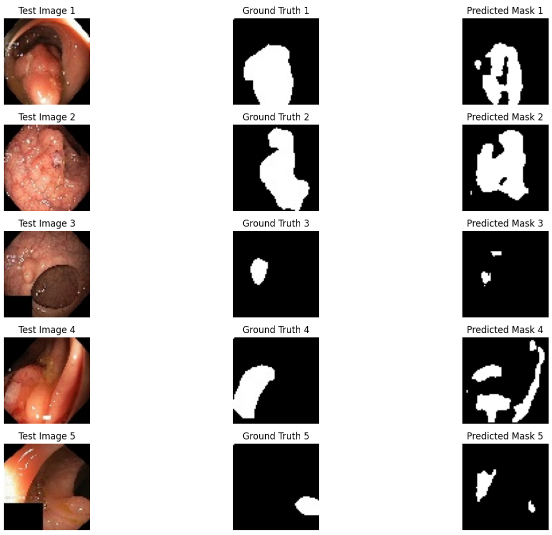

Fig. 11. Five random sample of the test image along side the ground truth mask and followed by the prediction of the model learned based on combination of IOU and DICE loss

Additionaly, the performance on 20% least confident test set, which Table IV shows the performance of model for each loss on these images .Figure 9 indicates four least confidant samples of the test image along side the ground truth mask and followed by the prediction of the model learned based on combination of IOU and DICE loss.

TABLE IV. RESIZED MODEL PERFORMANCE USING DIFFERENT LOSS FUNCTIONS ON 20% LEAST CONFIDENT TEST SET

Models	**Metrics**				
	Acc	***mIOU***	***mDice***	***Conf mean***	***Conf std***
IOU _DICE	88.67%	21.7%	32.7 %	0.88	0.051

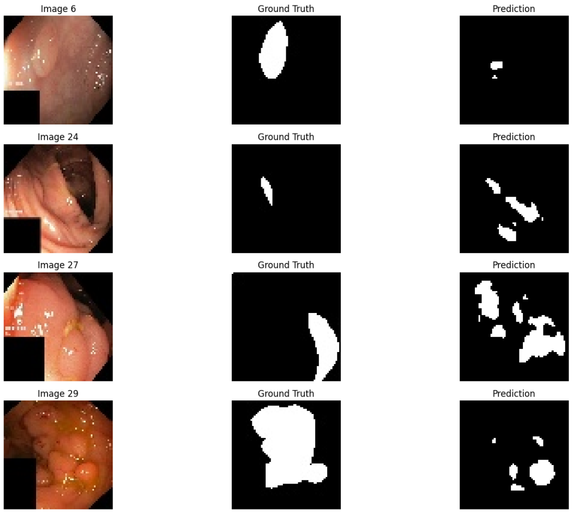

Fig. 12. Four least confidant samples of the test image along side the ground truth mask and followed by the prediction of the model learned based on combination of IOU and DICE loss

As explained in the method section to address the low performance on the bottom 20% of low-confidence images, we augmented our training set and then evaluated the ‘performance on both the entire test set and the bottom 20% subset. The numbers for evaluation metrics and confidence of estimation shown in Table V.

TABLE V.

Model trained on Augmented data	**Metrics**				
	Acct	***mIOU***	***mDice***	***Conf mean***	***Conf std***
All Test set	84.8 %	50.7%	63.9 %	0.957	0.030
20% bottom	89%	40.4%	53.6 %	0.914	0.039

The performance of Deep UNet model as mentioned in table VI. Is about 2% on average better than the standard unet model even with data augmentation. Feeding the Augmented data, this model improved the performance about 10%, and for the low 20% confident images it was improved around 9% for the mIOU and 5% for the mDICE.

TABLE VI. DEEP UNET ARCHITECTURES

Model	**Metrics**		
	Acct	***mIOU***	***mDice***
Deep UNet _all	84.5 6 %	52.50 %	62.16%
Deep UNet _all_aug	84.7 7%	60.95 %	71.28%
Deep UNet _aug_20%	93%	49%	58%%

The Attention Model though results a significant improvement over all, 19% for mIOU and 22% for mDICE. feeding augmented data the improvement is around 17% for mIOU and 13% for mDICE. The performance for the low 20% images improved around 9% each. (Table VII)

The pre-trained models didn’t result in any improved performance in comparison with the attention model, However, they are still better than the deep U-Net. Resnet50 has better metrics than MobileNetV2. Table VIII

TABLE VII. ATTENTION ARCHITECTURES

Model	Metrics		
	Acct	*mIOU*	*mDice*
Attention_UNet _all	85.63%	63.70 %	72.25 %
Attention_UNet _all_aug	86.14%	67.12 %%	76.32 %
Attention_UNet_Aug_ 20%	99.03%	51.24 %	59.53 %

TABLE VIII. FINE-TUNED PRE-TRAINED MODEL

Model	Metrics		
	Acct	*mIOU*	*mDice*
MobileNet	84.73%	57.19 %	67.71 %
ResNet50	85.73%	61.34 %	70.79 %

IV. CONCLUSION

The accuracy of the model using preprocessed images as input improved compared to using no preprocessing. Among the methods, morphological opening and closing yielded better performance, which is expected for medical images, as these preprocessing steps are generally helpful. However, the IoU (Intersection over Union) and Dice metrics, which measure the overlap between the predicted and actual segmentation, remained relatively low—especially for the bottom 20% of low-confidence images.

To address this, we experimented with different loss functions. The results showed that using a combination of weighted IoU loss and Dice loss significantly improved performance while maintaining model confidence.

To further tackle the issue of missing polyps, we augmented the test set by adding transformed versions of the original test images. This resulted in a 2% increase in Dice and IoU metrics across the entire test set, and a 20% improvement for the bottom 20% of low-confidence test images, which is promising.

The experimental results clearly demonstrate the impact of both architectural enhancements and data augmentation and combined loss function strategies on segmentation performance. The Deep U-Net model showed a modest but consistent improvement of approximately 2% over the standard U-Net, even when both were trained on augmented data, indicating that deeper architectures offer enhanced feature extraction capabilities.

Notably, feeding augmented data into the Deep U-Net led to a substantial performance gain of around 10%, with even more pronounced benefits observed on the most challenging 20% of the images (low-confidence cases), where improvements reached 9% in mIoU and 5% in mDice.

In contrast, the Attention U-Net architecture yielded the most significant gains overall, with improvements of 19% in mIoU and 22% in mDice, showcasing the effectiveness of attention mechanisms in enhancing spatial focus and feature relevance. Augmented training data further reinforced this performance, improving results by 17% in mIoU and 13% in mDice, and maintaining solid gains for the hardest examples (9% each for mIoU and mDice).

Interestingly, the use of pretrained encoder backbones (e.g., MobileNetV2 and ResNet50) did not outperform the Attention U-Net model, suggesting that while transfer learning provides strong initial features, custom attention-based architectures may better adapt to specific segmentation tasks—especially when trained end-to-end with task-specific data.

Although we managed to improve the IoU and Dice metrics across the full test set—particularly for the lowest-performing samples—we still aim to enhance the model architecture. For future work, several directions could be explored to further enhance segmentation performance and model robustness. Integrating attention mechanisms more deeply into pretrained backbones, such as ResNet or MobileNet, may yield better synergy between learned features and spatial focus. Additionally, applying more advanced data augmentation techniques or experimenting with transformer-based modules could improve the model's ability to capture long-range dependencies. Evaluating the models on larger, more diverse datasets and incorporating uncertainty estimation for active learning may also lead to better generalization and annotation efficiency.

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